Fiscal Year:	FY 2018	**Task Last Updated:**	FY 02/19/2019
PI Name:	Yodh, Arjun Ph.D.		
Project Title:	Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	06/10/2013	**End Date:**	06/09/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 6/9/2019 per NSSC information (Ed., 10/3/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AL27G		
Performance Goal No.:			
Performance Goal Text:			
	This proposal builds on and continues the research described in Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2) (Ed. note: see "Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2)"; grant NNX08AO80G in Task Book). The roots of the work derive from an experimental program that was peer-reviewed and accepted for flight on the International Space Station (ISS); the instrumentation at the heart of the original flight proposal for ISS was part of the so-called Light Microscopy Module (LMM). More recently, a Science Concept Review (SCR) for Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2) was held in June 2011, and the proposed research was strongly endorsed by the External Review Panel in their report on June 14 2011; thus, the science goals and experimental plans for the present proposal, Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3), have already been critically reviewed. The current research will be integrated into the hardware developed for the space-flights that are		

Task Description:	named by variants of the term “ACE” (e.g., ACE-H1). In addition, related complex fluid measurements have been carried out in the Binary Colloid Alloy Test (BCAT) BCAT-series flight experiments. Since the original proposal Low Volume Fraction Entropically Driven Colloidal Assembly was submitted and approved, the NASA budget has been in flux and soft matter science has evolved, too. In response to these developments, the science aims have evolved to take advantage of important recent breakthroughs in the field. Herein we describe these complex fluid experiments. Note also, whenever possible, we will be joining forces with our colleagues from Europe to perform collaborative experiments within the European Space Agency (ESA) and NASA facilities. The overall rationale for these investigations is that the elimination of particle sedimentation effects in microgravity creates a purely “thermodynamic” environment for colloidal suspensions wherein particle size, shape, and volume fraction, and interparticle interactions are the dominant determinants of the assembled structures. This feature enables us to make new materials that do not form on Earth and to elucidate the effects of gravity in soft matter science that are masked in ground-based experiments. The Phase 3 version of Low Volume Fraction Entropically Driven Colloidal Assembly explores a range of experimental complex fluids with themes related to and evolved from previous proposal aims. The original studies were centered around entropically driven low volume-fraction self-assembling complex fluids, suspensions composed of an ensemble of large particles, and additional smaller constituents (either a small colloidal particle, a polymer, or a micelle). The present studies retain many of the original themes but also have new focus that reflects the evolution of the complex fluid field, as well as new developments in our laboratory. Broadly, the proposed research will elucidate a wide-range of phenomena in condensed matter physics that has implications for phase-diagrams of entropically driven colloidal media, crystallization, and melting mechanisms in three-dimensional (3D) bulk media and in finite volumes, jamming, and glass formation in 3D, and, opportunistically, crystallization and melting mechanisms and aging of colloids/soft-matter based on more anisotropic particles and filaments on a variety of length scales. International collaborators are Prof. Luca Cipelletti, Université Montpellier II, Montpellier, France; Prof. Roberto Piazza, Department of Nuclear Engineering, Politecnico di Milano, Milan, Italy; Prof. Willem Kegel, Van 't Hoff Laboratory for Physical and Colloid Chemistry, Debye Research Institute, Utrecht University, Utrecht, The Netherlands. NOTE that project continues as "Crystal-to-Glass Transitions in Colloids (Flight Experiment)"; grant 80NSSC19K0348 ; Principal Investigator Arjun Yodh, Ph.D.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research can provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.
Task Progress:	[Ed. note February 2019: Below compiled from PI's annual report submitted May 2018] The abstracts that follow summarize the publications from June 2017 through February 2018; see Bibliography for citations and links to the publications: Davidson et al., Nature Communications, 2017 May 30: Drop drying and deposition phenomena reveal a rich interplay of fundamental scienceand engineering, give rise to fascinating everyday effects (coffee rings), and influence technologies ranging from printing to genotyping. Here we investigate evaporation dynamics, morphology, and deposition patterns of drying lyotropic chromonic liquid crystal droplets. These drops differ from typical evaporating colloidal drops primarily due to their concentration-dependent isotropic, nematic, and columnar phases. Phase separation occurs during evaporation, and in the process creates surface tension gradients and significant density and viscosity variation within the droplet. As a result, the drying multiphase drops exhibit different convective currents, drop morphologies, and deposition patterns (coffee-rings). Cubuk et al., Science, 2017 November: When deformed beyond their elastic limits, crystalline solids flow plastically via particle rearrangements localized around structural defects. Disordered solids also flow, but without obvious structural defects. We link structure to plasticity in disordered solids via a microscopic structural quantity,“softness,” designed by machine learning to be maximally predictive of rearrangements. Experimental results and computations enabled us to measure the spatial correlations and strain response of softness, as well as two measures of plasticity: the size of rearrangements and the yield strain. All four quantities maintained remarkable commonality in their values for disordered packings of objects ranging from atoms to grains, spanning seven orders of magnitude in diameter and 13 orders of magnitude in elastic modulus. These commonalities link the spatial correlations and strain response of softness to rearrangement size and yield strain, respectively. ED. NOTE: This project continues in December 2018 as "Crystal-to-Glass Transitions in Colloids (Flight Experiment)," grant 80NSSC19K0348, with the same Principal Investigator Arjun Yodh, Ph.D. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 06/16/2025)
Articles in Peer-reviewed Journals	Davidson ZS, Huang Y, Gross A, Martinez A, Still T, Zhou C, Collings PJ, Kamien RD, Yodh AG. "Deposition and drying dynamics of liquid crystal droplets." Nat Commun. 2017 May 30;8:15642. https://doi.org/10.1038/ncomms15642 ; PubMed PMID: 28555621; PubMed Central PMCID: PMC5460001 , May-2017
Articles in Peer-reviewed Journals	Cubuk ED, Ivancic RJS, Schoenholz SS, Strickland DJ, Basu A, Davidson ZS, Fontaine J, Hor JL, Huang YR, Jiang Y, Keim NC, Koshigan KD, Lefever JA, Liu T, Ma XG, Magagnosc DJ, Morrow E, Ortiz CP, Rieser JM, Shavit A, Still T, Xu Y, Zhang Y, Nordstrom KN, Arratia PE, Carpick RW, Durian DJ, Fakhraai Z, Jerolmack DJ, Lee D, Li J, Riggleman R, Turner KT, Yodh AG, Gianola DS, Liu AJ. "Structure-property relationships from universal signatures of plasticity in disordered solids." Science. 2017 Nov 24;358(6366):1033-7. https://doi.org/10.1126/science.aai8830 ; PubMed PMID: 29170231 ; PubMed Central PMCID: PMC6047528 , Nov-2017